

Ticket Systems

Learning Objectives

The learning objectives in the tutorial are:

- Installing a ticketing system on the cloud server
- Performing some basic configurations

Tools

- Znuny-6.2.1
- Rocky 8 Server x86_64

Contents

Tut 1: Tutorial (20 min)	1
Tut 2: Tutorial (10 min)	10

Tut 1: Tutorial (20 min)

Goal of this task is to set up Znuny. After this task, the following data should be available and stored somewhere, e.g., in a local editor, for easy access:

- `< frontendfloatingip >` - the floating ip of the frontend server
- `< otrsuserpassword >` - the password of user otrs
- `< dbuserpassword >` - the password of the database user
- `< agent >` - Znuny agent user name
- `< agentpassword >` - Znuny agent password
- `< root >` - Znuny root user
- `< rootpassword >` - Znuny root password

The `< frontendfloatingip >` can be found in the cloud administration tool cloud.gwdg.de

Install the basics:

```
$ sudo dnf install -y epel-release httpd mariadb mariadb-server
$ sudo dnf install -y cpanminus gcc dnf-plugins-core
$ sudo yum config-manager --set-enabled powertools $ sudo yum install -y nano
```

Install Znuny via RPM:

```
$ link=https://download.znuny.org/releases/RPMS/rhel/7/znuny-6.2.1-01.noarch.rpm
$ sudo yum install -y $link
```

Install required Perl Modules (i.e., complete the missing ones):

Check the underline (.) in the text

```
$ sudo yum install -y "perl(Moo)" "perl(Text::CSV_XS)" "perl(YAML::XS)"
$ sudo yum install -y "perl(ModPerl::Util)" "perl(Mail::IMAPClient)" "perl(JSON::XS)"
$ sudo yum install -y "perl(Encode::HanExtra)" "perl(Crypt::Eksblowfish::Bcrypt)"
$ sudo cpanm JavaScript::Minifier::XS CSS::Minifier::XS
```

MySQL / Maria DB needs some config modifications.

Create a new file with a text editor of your choice, e.g. nano:

Check the underline (.) in the text

```
$ sudo nano /etc/my.cnf.d/znuny_config.cnf
```

Add following content:

Check the underline (.) in the text

```
[client]
max_allowed_packet=256M
[mysqld]
innodb_file_per_table
innodb_log_file_size=256M
max_allowed_packet=256M
```

Restart the MariaDB database to apply the changes:

```
$ sudo systemctl start mariadb
```

Configure the webserver: The Apache config is already in place as you have used RPM install previously.

Enable MPM prefork module:

```
$ sudo nano /etc/httpd/conf.modules.d/00-mpm.conf
Comment "LoadModule mpm_event_module modules/mod_mpm_event.so"
Uncomment "LoadModule mpm_prefork_module modules/mod_mpm_prefork.so"
```

Znuny uses port 80 by default. You might want to check if this port is already in use using the following command and free it if necessary:

```
$ sudo netstat -tulpn | grep LISTEN
```

Once done, restart the web server to apply the changes:

```
$ sudo systemctl restart httpd
```

Enable the web server and the database to start on boot:

```
$ sudo systemctl enable mariadb httpd
```

Add otrs user to sudoers and set password < otrsuserpassword >:

```
$ sudo usermod -aG wheel otrs
$ sudo passwd otrs
```

Enable Znuny Cron, use *<otrsuserpassword>* for login:

```
$ su - otrs  
$ bin/Cron.sh start
```

Set up the database:

Log in to MySQL console as database admin user. Ensure to replace `<dbuserpassword>` in the listing below with a password of your liking and note it.

```
$ sudo mysql -uroot
```

Create a database:

```
$ CREATE DATABASE otrs CHARACTER SET utf8;
```

Create a database user (i.e, otrs), and its password ;dbuserpassword;

You need to replace the ;dbuserpassword; with 'yourpassword'

```
$ CREATE USER otrs@localhost IDENTIFIED BY <dbuserpassword>;
```

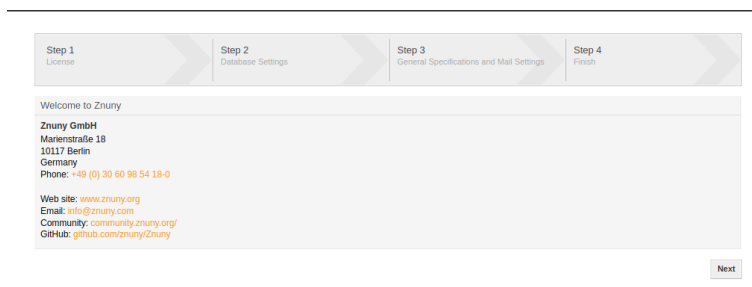
Assign user privileges to the new database and quit:

```
$ GRANT ALL PRIVILEGES ON otrs.* TO otrs@localhost;  
$ FLUSH PRIVILEGES;  
$ quit
```

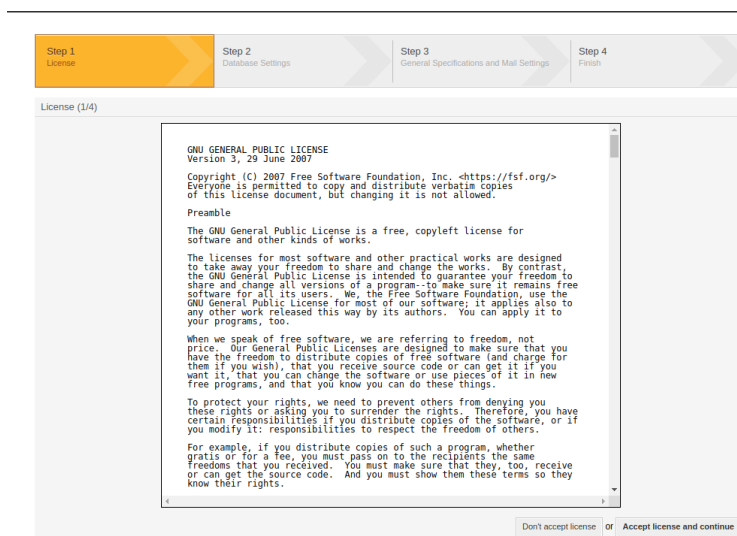
Visit `<frontendfloatingip>/otrs/installer.pl` and follow the configuration steps:

Steps

1. Click next



2. Accept license, and continue



3. Type: MySQL, install type: Use an existing database for OTRS, and click next

Step 1 License

Step 2 Database Settings

Step 3 General Specifications and Mail Settings

Step 4 Finish

Database Selection (2/4)

Type: MySQL
Oracle
PostgreSQL

Install Type: ☐ Create a new database for OTRS
☒ Use an existing database for OTRS

Next

4. Configure MySQL, user: otrs, Password: < dbuserpassword >, host: 127.0.0.1, Database name: otrs, and click on database check. If it says successful then click next

Step 1 License

Step 2 Database Settings

Step 3 General Specifications and Mail Settings

Step 4 Finish

Configure MySQL (2/4)

User: otrs

Password: *****
Enter the password for the database user.

Host: 127.0.0.1

Database name: otrs

Result of database check
✓ Database check successful.

Back Next

5. Database setup successful!. Click next

Step 1 License

Step 2 Database Settings

Step 3 General Specifications and Mail Settings

Step 4 Finish

Create Database (2/4)

✓ Processing otrs-schema
✓ Processing otrs-initial_insert
✓ Processing post statements
Database setup successful!

Next

6. System Settings, no change is required click next.

Step 1
License

Step 2
Database Settings

Step 3
General Specifications and Mail Settings

Step 4
Finish

System Settings (3/4)

SystemID: 19 ▼
The identifier of the system. Each ticket number and each HTTP session ID contain this number.

System FQDN: znuny.novalocal
Fully qualified domain name of your system.

AdminEmail: support@yourhost.example.com
Email address of the system administrator.

Organization: Example Company

Log
LogModule: Syslog ▼
Log backend to use.

Webfrontend
Default language: English (United States) ▼
Default language.

CheckMXRecord: Yes ▼
Email addresses that are manually entered are checked against the MX records found in DNS. Don't use this option if your DNS is slow or does not resolve public addresses.

Next

7. Mail Configuration. Skip this step

Step 1
License

Step 2
Database Settings

Step 3
General Specifications and Mail Settings

Step 4
Finish

Mail Configuration (3/4)

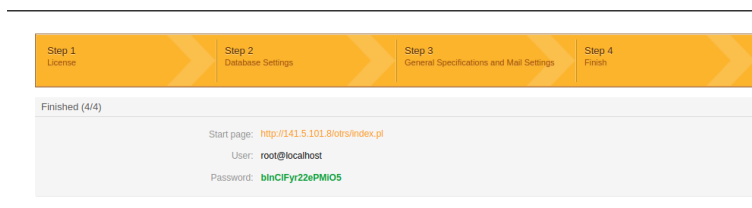
Configure Outbound Mail
Outbound mail type: Sendmail ▼
Select outbound mail type.
Outbound mail port: 25
Select outbound mail port.

Configure Inbound Mail
Inbound mail type: IMAP ▼
Select inbound mail type.
Inbound mail host:
Inbound mail host.
Inbound mail user:
User for inbound mail.
Inbound mail password:
Password for inbound mail.

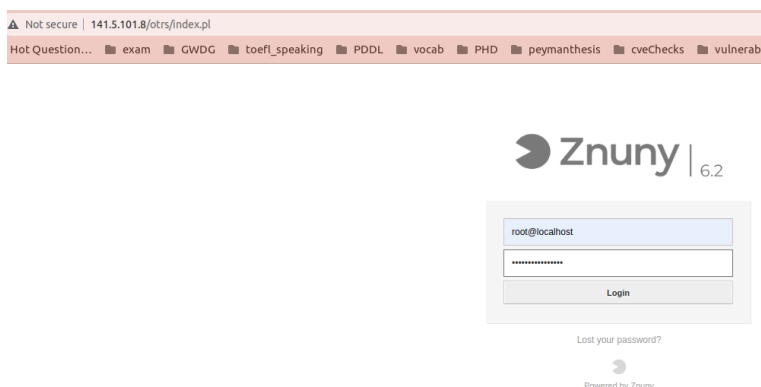
Check mail configuration or Skip this step



8. Finished, copy the password `<rootpassword>` and username `<root>`. Open the given link in new tab.



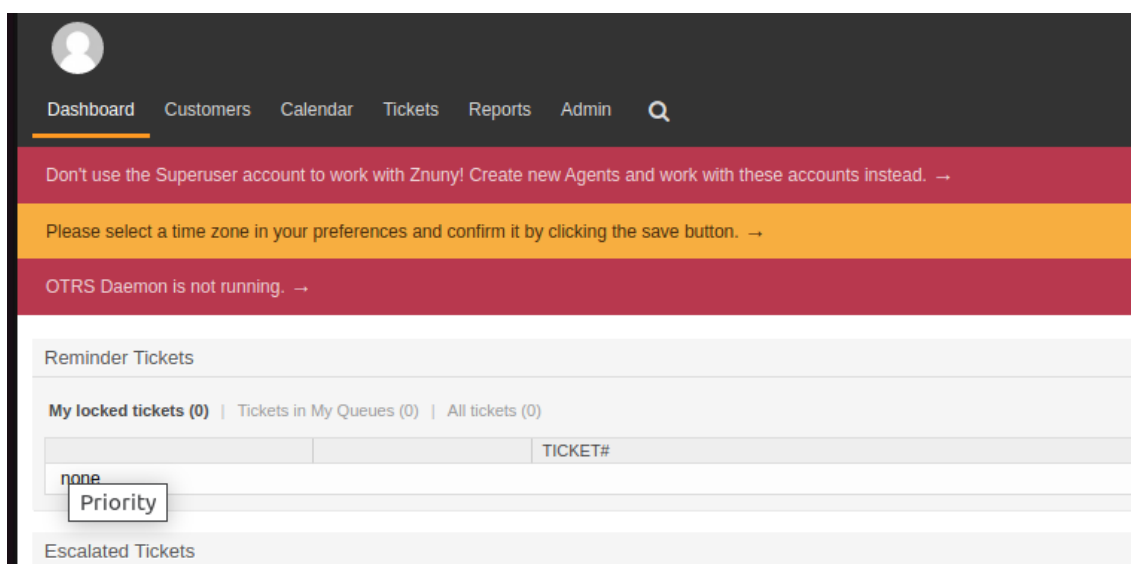
9. Agent login page. Enter the username and password from last step, and click on login. Please note that your IP address will differ from the one in the picture.



Next, configure the system.

The first page you see after login is this. Consider the **Red and Yellow/Orange notification** at the top of the page. We need to first resolve those issues by clicking on them:

- OTRS Daemon is not running.
- Time zone
- Do not use Super User account.



1. OTRS Daemon is not running

Please ensure you are still using otrs user:

```
$ /opt/otrs/bin/otrs.Daemon.pl stop  
$ /opt/otrs/bin/otrs.Daemon.pl start
```

2. Please select a time zone (Berlin) in your preferences and confirm it by clicking the save button (checkmark at the end of the line).

The screenshot shows the Znuzy 6.2 user interface. At the top, there is a navigation bar with links to Dashboard, Customers, Calendar, Tickets, Reports, and Admin. A search icon is also present. Below the navigation bar, there is a red banner with the text: "Don't use the Superuser account to work with Znuzy! Create new Agents and work with these accounts instead. --". Below this, there is an orange banner with the text: "Please select a time zone in your preferences and confirm it by clicking the save button. --". The main content area is titled "Personal Preferences" and "User Profile". It contains several sections: "Change password" with fields for Current password, New password, and Verify password; "Avatar" with a note about changing the avatar image; "Language" with a dropdown menu set to "English (United States)"; "Time Zone" with a dropdown menu set to "Europe/Berlin"; and "Out Of Office Time" with fields for Active status, Start date, and End date. Each section has a checkmark icon on the right to save changes.

3. Don't use the Superuser account to work with Znuny! Create new Agents and work with these accounts instead

Don't use the Superuser account to work with Znuny! Create new Agents and work with these accounts instead. --

Agent Management

Actions

Wildcards like "*" are allowed.

Add Agent

Hint

Agents will be needed to handle tickets.
Attention: Don't forget to add a new agent to groups and/or roles!

List (1 total)

USERNAME	NAME	EMAIL	LAST LOGIN	VALIDITY	CHANGED	CREATED
root@localhost	Admin OTRS	root@localhost	02/16/2023 07:48 (Europe/Berlin)	valid	02/15/2023 14:59 (Europe/Berlin)	02/15/2023 14:59 (Europe/Berlin)

Click on add agent, and fill out the form. Make sure to take note of the login credentials *< agent >* and *< agentpassword >*.

Don't use the Superuser account to work with Znuny! Create new Agents and work with these accounts instead. --

Agent Management Add Agent

Actions

Go to overview

Hint

Agents will be needed to handle tickets.
Attention: Don't forget to add a new agent to groups and/or roles!

Add Agent

Title or salutation: admin

★ Firstname: admin

★ Lastname: admin

★ Username: admin

Password: ****

Will be auto-generated if left empty.

★ Email: znuny_test@outlook.com

Mobile:

Validity: valid

Save or Cancel

After clicking on save, the next page is asking about the permission setting. We give it full permission by selecting all.

Don't use the Superuser account to work with Znuny! Create new Agents and work with these accounts instead. --

Manage Agent-Group Relations Change Group Relations for Agent admin admin

Actions

Go to overview

Filter

Just start typing to filter...

Change Group Relations for Agent admin admin

GROUP	RO	MOVE INTO	CREATE	NOTE	OWNER	PRIORITY	RW
admin	☒	☒	☒	☒	☒	☒	☒
stats	☐	☐	☐	☐	☐	☐	☐
users	☐	☐	☐	☐	☐	☐	☐

Save or Save and finish or Cancel

Reference

ro Read only access to the ticket in this group/queue.

move_into Permissions to move tickets into this group/queue.

create Permissions to create tickets in this group/queue.

note Permissions to add notes to tickets in this group/queue.

owner Permissions to change the owner of tickets in this group/queue.

priority Permissions to change the ticket priority in this group/queue.

rw Full read and write access to the tickets in this group/queue.

Later for the exercise section, we should logout (by clicking on the profile picture), and login with the new agent you have just created.

Tut 2: Tutorial (10 min)

This is a short description of the features of the Znuzy web interface. You may use this as reference for the exercise.

Agent Web Interface: Overview

The Agent Web Interface consists of the following tabs

- Dashboard, Customers, Calendar, Tickets, Reports, Admin, and Search
 1. The **Dashboard** is the main page of the system, here you can get an overview about the tickets and other stuff related to the ticket activity.
 2. **Tickets** button, you will get a sub-menu with options to change the ticket's view, create a new ticket or search for a specific ticket.
 3. The **Report** (Statistics) button presents a menu that allows you to choose from an overview of the registered statistics, creating a new one or importing an existing one.
 4. The **Customers** button leads you to the Customer Management screen.
 5. By clicking the **Admin** button, you can access all of the administrator modules, which allows you to create new agents, queues, etc.
 6. There is also a **Search** button to make ticket searches.

Agent Web Interface: Dashboard

- Dashboard's widgets
 - Reminder Tickets (Tickets that are set as pending and the reminder date has been reached.)
 - Escalated Tickets (Tickets that are escalated.)
 - New Tickets (Tickets that have state New.)
 - Open Tickets / Need to be answered (Tickets that have state Open and are ready for work with them.)
- Dashboard Settings
 - It allows you to control the widgets you want to show or hide.

Agent Web Interface: Admin

In the Admin page of the OTRS web interface, an agent can add/modify agents, customers and queues, ticket and mail settings, install additional packages such as FAQ and ITSM, and much more.

- Agent Management
 - Administrators can add, change or deactivate agent accounts. Furthermore, they can also manage agent preferences, including the language and notification settings for the individual agent's interface.
- Ticket Settings
 - The statuses (predefined states) of tickets, different queues, the template for responding to a specific ticket, and so on are specified or set here.

Agent Web Interface: Admin

- Users, Groups and Roles

-
- OTRS supports different types of users and customers. In this section you can create or modify new users, groups, customers, and etc.

- Communication and Notifications

In this section, in "PostMaster Mail Accounts", you can configure the inbound and outbound mail setting. And in "Email Addresses" section, you can set the email for system.